Informatique quantique 2

Quantum information Reminder

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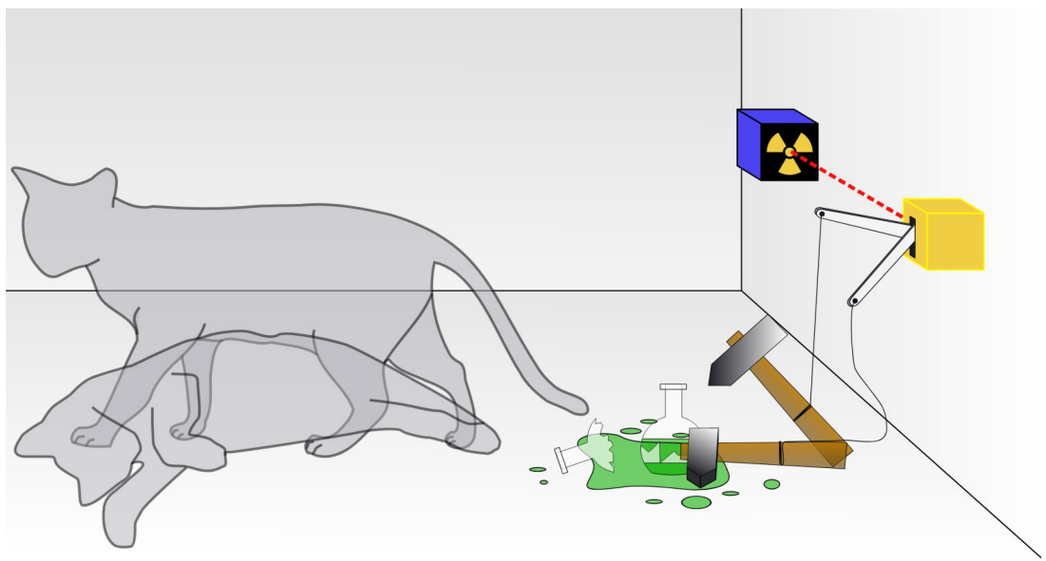
2025-2026

Notations

Notations

Qubit :

Schrödinger cat



$$|\psi\rangle = \alpha\,|dead\rangle + \beta\,|alive\rangle$$

Schrödinger plates

$$|\psi\rangle = \alpha\,|intact\rangle + \beta\,|broken\rangle$$

$$\beta \approx 1$$

Notations

Qubit :

- Any qubit that is not null can be in a « **pure state** »:

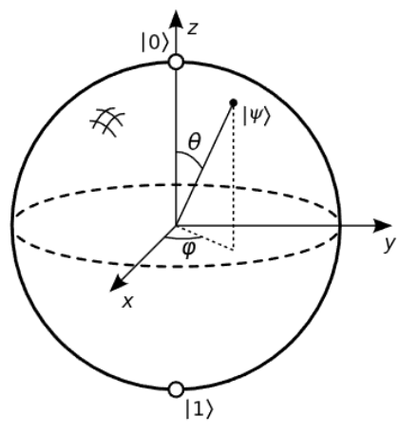
$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \qquad \alpha, \beta \in \mathbb{C}$$

psi alpha beta theta

is equivalent to :

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

This state can be represented on a Bloch sphere :



$$0 < \theta < \pi$$
$$-\pi < \phi < \pi$$

*At the beginning we had 4 degrees of freedom : $Re(\alpha), Im(\alpha), Re(\beta), Im(\beta)$.
We loose 2 degree due to normalization constraint and the global phase

* With this visualization, only θ determines the probability of getting state 0 or state 1

Dirac notations:

$|...\rangle$: a ket

$\langle...|$: a bra

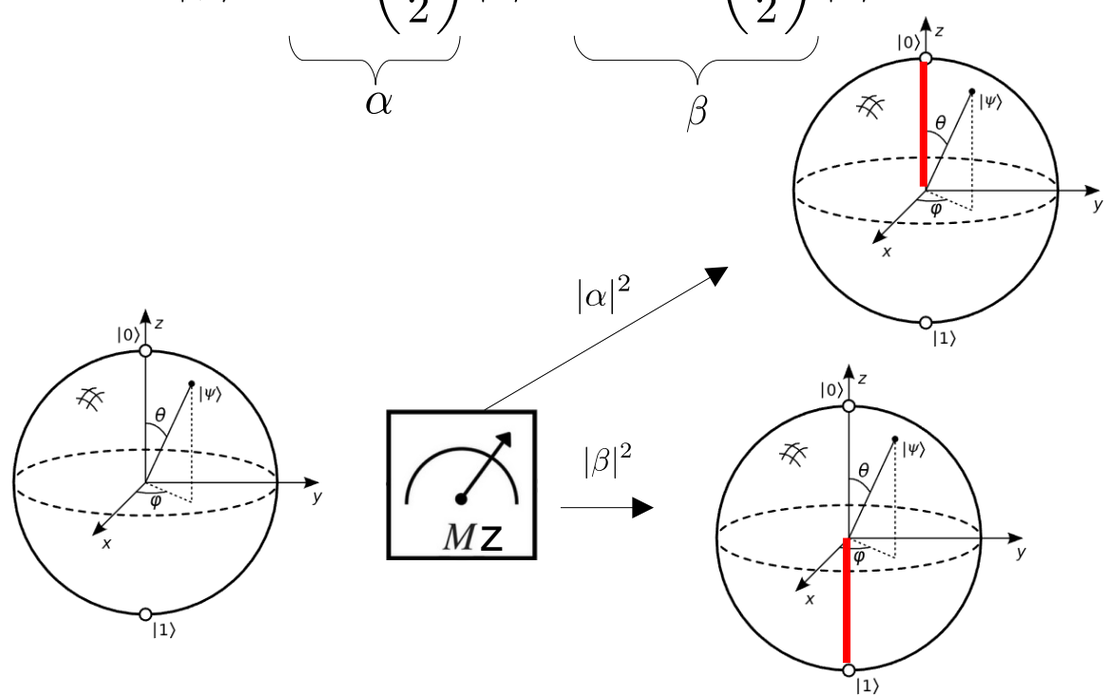
$\langle...|...\rangle$: a bra-ket

Notations

<u>Measurement :</u>

- The measurement is a projection of the quantum state in a classical state.

$$|\psi\rangle = \underbrace{\cos\left(\frac{\theta}{2}\right)}_{\alpha}|0\rangle + \underbrace{e^{i\phi}\sin\left(\frac{\theta}{2}\right)}_{\beta}|1\rangle$$



This is the measurement that fixes the final state.

Notations

Measurement basis:

- Usual basis to do measurements :

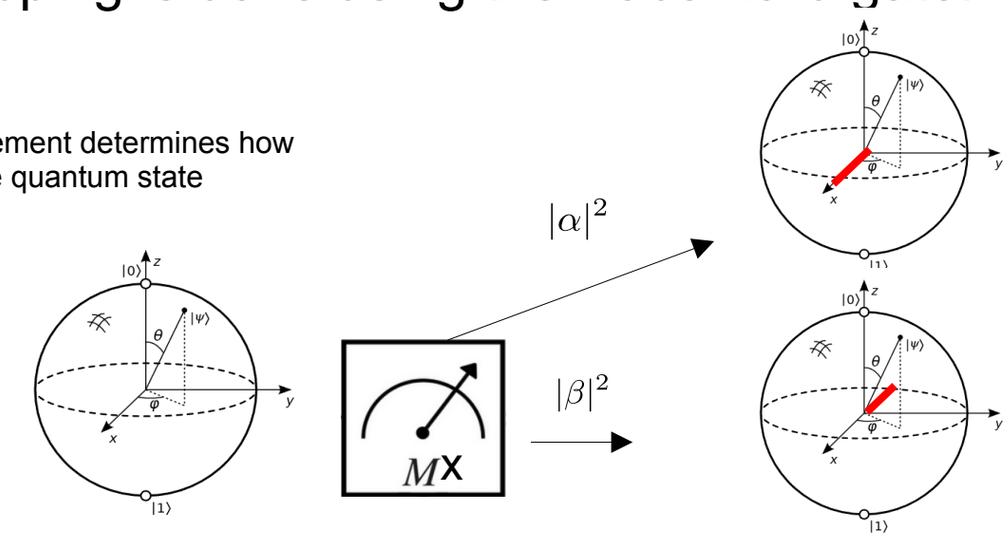
$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Another basis example :

$$|0\rangle \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle, |1\rangle \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$$

This mapping is done using the Hadamard gate.

*The measurement determines how we project the quantum state



To proove that the vectors constitute A basis:

- All the vector are orthogonal to each other :

$$\langle 0|1\rangle = 0$$

- The norm of the vectors is 1 :

$$|\,|0\rangle\,| = \sqrt{x^2 + y^2} = \sqrt{1^2 + 0^2} = 1$$

- The number of vector is the same as the number of dimensions

$$2 \text{ in } \mathbb{C}^2$$

Notations

<u>Representation of many qubits:</u>

- A quantum state of many can be represented with a **tensor product**

- Tensor product of vectors :

$$\vec{a} = \begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2 \qquad \vec{b} = \begin{pmatrix} c \\ d \end{pmatrix} \in \mathbb{R}^2$$

$$\vec{a} \otimes \vec{b} = \begin{pmatrix} a \times \vec{b} \\ b \times \vec{b} \end{pmatrix} = \begin{pmatrix} a \times \begin{pmatrix} c \\ d \end{pmatrix} \\ b \times \begin{pmatrix} c \\ d \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}$$

$$\vec{a} \otimes \vec{b} \in \mathbb{R}^4$$

Notations

<u>Example of quantum states with 2 qubits:</u>

$$|00\rangle = |0\rangle\,|0\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix}$$

$$|01\rangle = |0\rangle\,|1\rangle = |0\rangle \otimes |1\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix}$$

$$|10\rangle = ...$$

$$|11\rangle = ...$$

- A 2-qubit quantum state :

$$|\phi\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|10\rangle + \delta\,|10\rangle$$

- The norm also applies to many-qubit states :

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$$

Notations

<u>Matrices:</u>

Number of lines

$$A^{n\times n} = \begin{array}{c} a_{ij} \\ \\ \\ \end{array} \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{pmatrix}$$

Number of columns

j vary

i vary

Notations

<u>Matrices product:</u>

$$A^{2\times 2} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$B^{2\times 2} = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$$

$$\begin{array}{cc} & \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} & \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix} \end{array}$$

$$AB = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix}$$

The number of columns in A should be equal to the number of lines in B

Notations

Matrix transpose:

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \\ \vdots & \vdots \\ a_{n,1} & a_{n,2} \end{pmatrix} \ , \ A^\top = \begin{pmatrix} a_{1,1} & a_{2,1} & \dots & a_{n,1} \\ a_{1,2} & a_{2,2} & \dots & a_{n,2} \end{pmatrix}$$

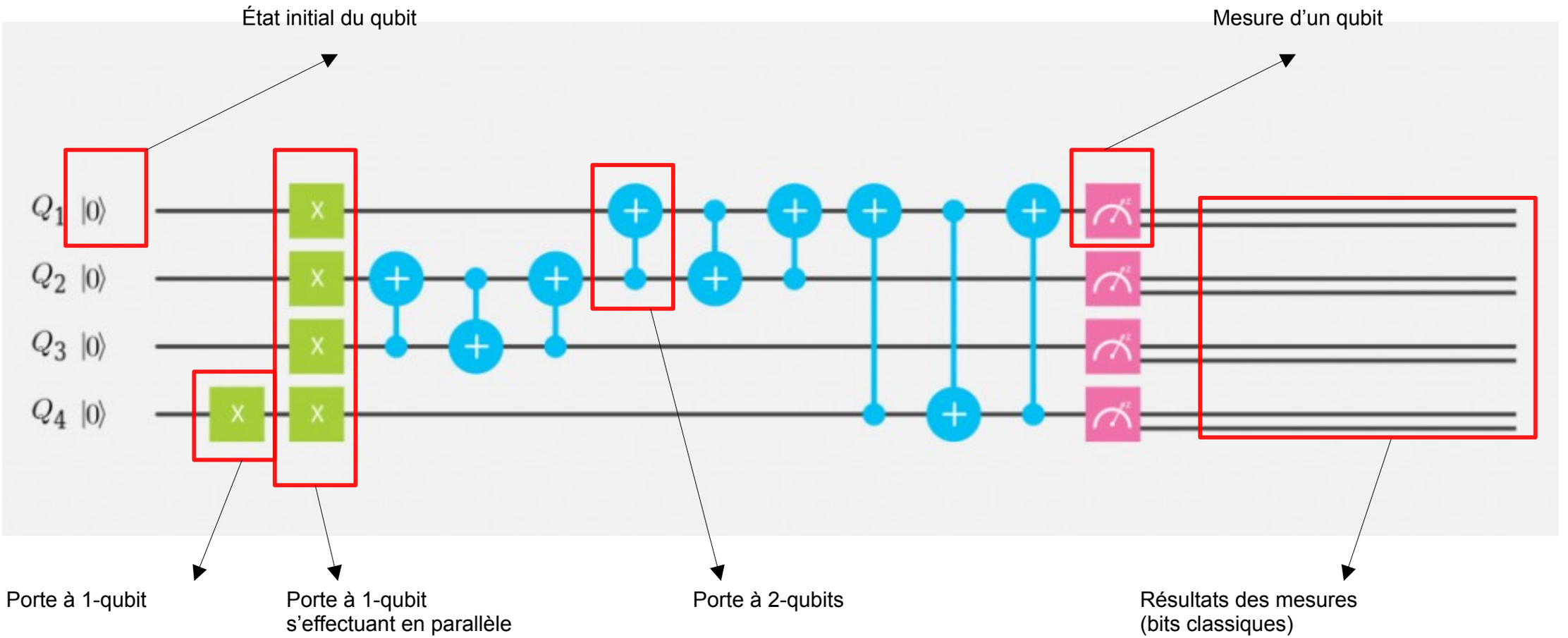
Conjugate transpose matrix :

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \dots & a_{n,n} \end{pmatrix} \ , \ \bar{A}^\top = A^{*\top} = A^\dagger = \begin{pmatrix} a_{1,1}^* & a_{2,1}^* & \dots & a_{n,1}^* \\ a_{1,2}^* & a_{2,2}^* & \dots & a_{n,2}^* \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n}^* & a_{2,n}^* & \dots & a_{n,n}^* \end{pmatrix}$$

Quantum processes

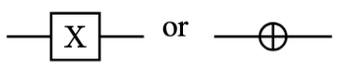
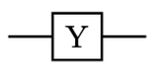
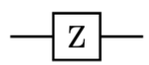
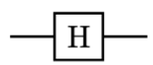
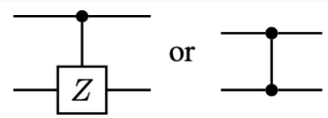
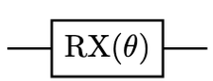
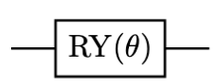
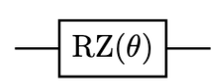
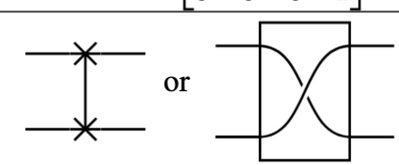
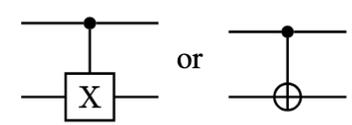
Quantum processes

Quantum circuit:



Quantum processes

Quantum operations:

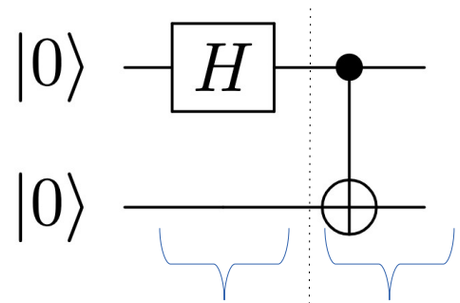
Quantum gate	Pauli-X (X)	Pauli-Y (Y)	Pauli-Z (Z)	Hadamard (H)	Controlled-Z (CZ)
Mathematical form	$\mathrm{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\mathrm{Y} = \begin{bmatrix} 0 & -\iota \\ \iota & 0 \end{bmatrix}$	$\mathrm{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\mathrm{H} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$\mathrm{CZ} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
Graph representation					
Quantum gate	Rotational single-qubit gate along X-axis ($\mathrm{RX}(\theta)$)	Rotational single-qubit gate along Y-axis ($\mathrm{RY}(\theta)$)	Rotational single-qubit gate along Z-axis ($\mathrm{RZ}(\theta)$)	SWAP	Controlled-NOT (CNOT, CX)
Mathematical form	$\mathrm{RX}(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\iota\sin(\frac{\theta}{2}) \\ -\iota\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}$	$\mathrm{RY}(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}$	$\mathrm{RZ} = \begin{bmatrix} e^{-\iota\frac{\theta}{2}} & 0 \\ 0 & e^{\iota\frac{\theta}{2}} \end{bmatrix}$	$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\mathrm{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Graph representation					

Source: https://www.researchgate.net/publication/352280311_The_dilemma_of_quantum_neural_networks

Quantum processes

<u>Entanglement:</u> Circuit for creating a Bell pair

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



Exercise: Check that the circuit gives the Bell pair

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \otimes \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \qquad \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 1 & 0\\ 0 & 1 & 0 & 1\\ 1 & 0 & -1 & 0\\ 0 & 1 & 0 & -1\end{pmatrix}$$

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \qquad CNOT = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix}$$

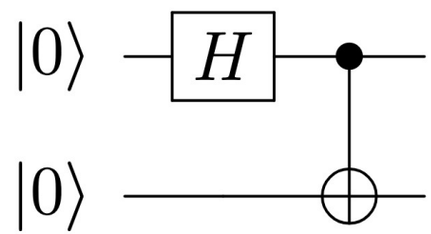
Detail of the computation :

$$(\mathrm{CNOT})(\mathrm{H}\otimes\mathrm{I})\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 1 & 0\\ 0 & 1 & 0 & 1\\ 1 & 0 & -1 & 0\\ 0 & 1 & 0 & -1\end{pmatrix}\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\\0\end{pmatrix} = \begin{pmatrix}1/\sqrt{2}\\0\\0\\1/\sqrt{2}\end{pmatrix}$$

Quantum processes

<u>Entanglement:</u> Circuit for creating a Bell pair

$$|\psi_{Bell}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



This state is particularly interesting because measuring the first qubit gives you **instantly** information on the second qubit (even if it's a million km away).

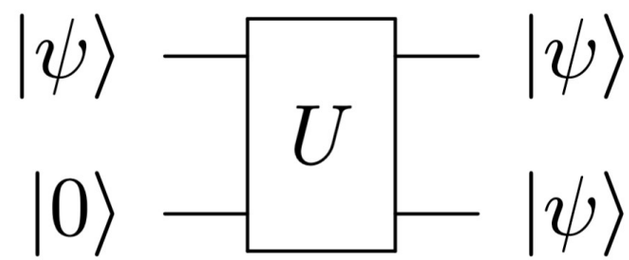
This state **cannot be factored** into a tensor product such that :

$$|\psi_{Bell}\rangle = |x\rangle \otimes |y\rangle$$

This state is **entangled.**
Property specific to quantum mechanics (does not exist in classical physics)

Quantum processes

<u>No cloning theorem:</u>
It is impossible to perfectly clone a quantum state such that :



This kind of operation would require a transformation that is not linear (hence not unitary).

The only way to make a perfect copy of $|\psi\rangle$ is to prepare it several time.

<u>Issue:</u> measurement destroys the quantum state. It can be difficult to correct errors happening on a quantum state.

Quantum processes

<u>Properties of quantum operations:</u>

Quantum operations are unitary matrices (matrix with complex coefficients). Unitaries verify :

$$U^\dagger U = U U^\dagger = I$$

This property implies that : $U^{-1} = U^\dagger$

So in general it means that any operation $U\left|\psi\right\rangle \rightarrow \left|\psi'\right\rangle$ can be later reversed using the matrix $U^\dagger$

=> Unitary information cannot destroy quantum information.

Quantum processes

<u>Properties of quantum operations:</u>

We can write the unitary as a matrix exponential (this will be proven subsequently):

$$U = e^{iH}$$

Where H is a hermitian matrix. Hermitian just means that the matrix is equal to its self adjoint :

$$H = H^{\dagger}$$

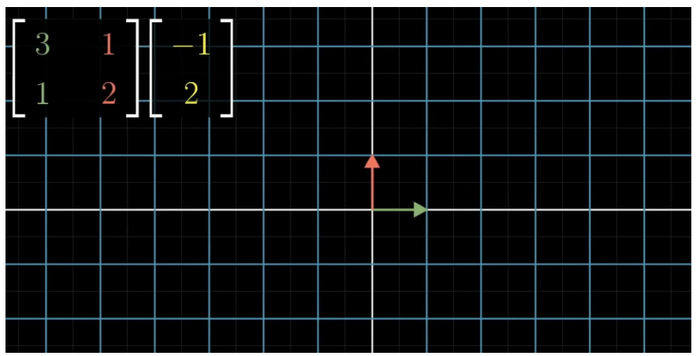
Some matrices are both Hermitian and unitaries. This is the case of Pauli matrices :

$$\sigma^x = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma^y = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \sigma^z = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

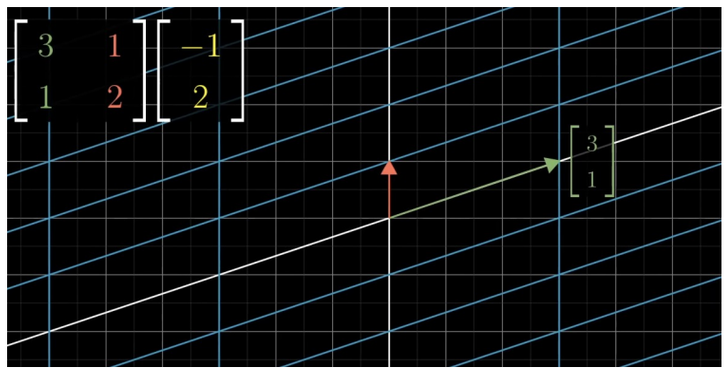
Quantum processes

<u>Recall on eigenvalue and eigenvectors:</u>
Matrices can be viewed as a basis change :

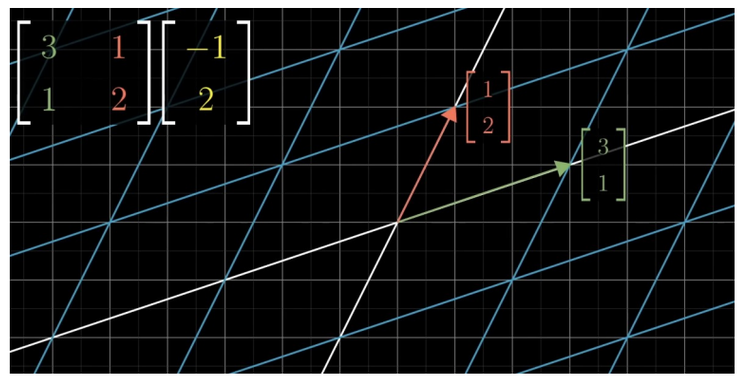
Initial basis



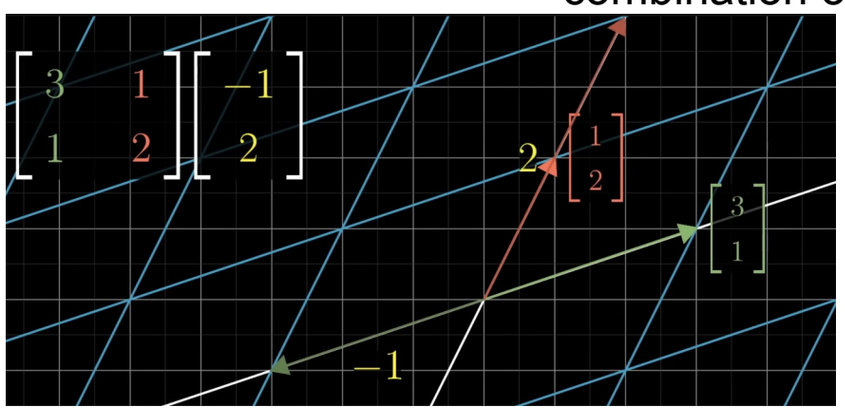
transform the X axis

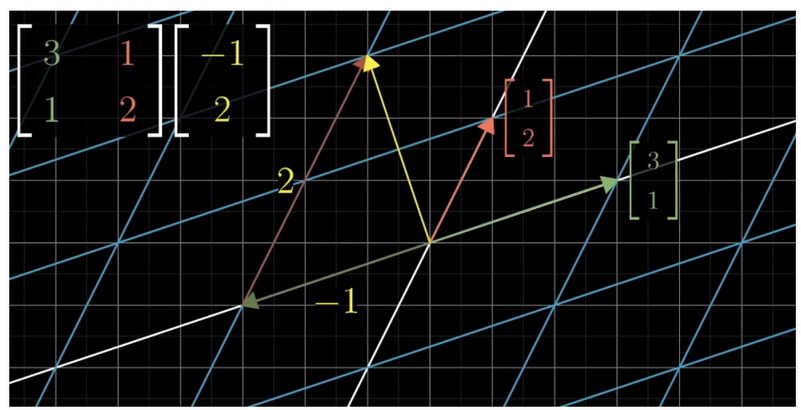


transform the Y axis



The result of matrix multiplication is the linear combination of the new basis vector





Source: https://www.youtube.com/watch?v=kYB8IZa5AuE&list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab&index=3

Quantum processes

Spectral theorem:

If we have a hermitian matrix $\mathcal{H}$, there exists an orthonormal basis consisting only of eigenvectors of $\mathcal{H}$ (eigenvectors that are linearly independent).

We can then use the eigenvalue equation to find the eigenvectors and associated eigenvalues:

$$\mathcal{H} \left| v_i \right\rangle = \lambda_i \left| v_i \right\rangle$$

Why is this important ?

- Because it gives you the **possible outcomes** when measuring a quantum state. Measurements are described by Hermitian operators.
- The **eigenvalues** and **eigenvectors** describe the possible outcomes of a measurement

Quantum processes

<u>Example with z-basis measurements :</u>

Let's assume we want to measure the quantum system in the z-basis.
The Z operator corresponds to :

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Is this operator Hermitian ?

$$Z^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^{*\top} = \begin{pmatrix} 1^* & 0^* \\ 0^* & -1^* \end{pmatrix}^\top = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

=> Yes it is Hermitian (in fact it is also a unitary)

What are the eigenvectors and eigenvalues of the Z operator ?
→ it is diagonal so the eigenvalues are in the diagonal

$$Z\,|0\rangle = \underbrace{+1}_{\text{eigenvalue}}\,\underbrace{|0\rangle}_{\text{eigenvector}} \qquad Z\,|1\rangle = \underbrace{-1}_{\text{eigenvalue}}\,\underbrace{|1\rangle}_{\text{eigenvector}}$$

Thank you for your attention !